

PolyVRGeo: Immersive Engineering Toolkit for Deep Geothermal Energy

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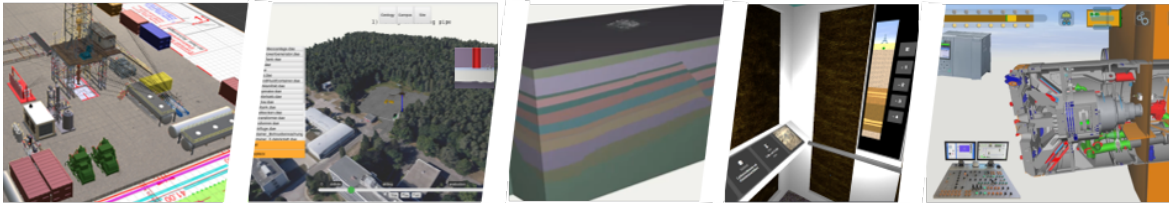


Figure 1: PolyVRGeo Use Cases: Drill Site Configurator, Underground Visualization and Tunnel Boring Machine Simulator

ABSTRACT

Geothermal energy offers a sustainable and reliable energy source. However, VR engineering methods have seen limited application in this field. This work introduces an Immersive Engineering Toolkit, which leverages an open-source VR engine to address challenges across the geothermal lifecycle, such as drill site planning, subsurface visualization, engineering simulations, and public engagement. By integrating heterogeneous data formats and enabling collaborative environments, the toolkit aims to enhance cost-efficiency, safety, decision-making and stakeholder communication. By unifying scientific, engineering, and societal aspects, the toolkit aims to provide a holistic approach to accelerate the adoption of deep geothermal energy systems.

Index Terms: Virtual reality, immersive visualization, deep geothermal energy, heterogeneous data, simulation, collaboration.

1 INTRODUCTION

Geothermal energy is a vital component in the transition to renewable energy sources. With its low carbon footprint and the ability to provide baseload power, geothermal energy contributes significantly to climate goals, particularly in urban areas where efficient energy storage and distribution are critical. However, the deployment of deep geothermal energy systems involves complex processes, from subsurface exploration to drilling and operation, which require interdisciplinary collaboration and clear communication among stakeholders. This complexity often leads to challenges in planning, decision-making, and public acceptance, underscoring the need for innovative tools to streamline these processes. Virtual reality (VR) and virtual engineering methods, widely used in industries such as automotive, aerospace, and marine engineering, could offer transformative potential for the geothermal sector. These technologies enable immersive, interactive, and realistic visualizations of complex systems, supporting decision-making, minimizing er-

rors, and reducing costs by simulating scenarios before implementation.

However, virtual reality and virtual engineering are rarely utilized in geothermal energy, with limited use cases explored by research institutes and rarely adopted by industry. The most relevant work is presented by Rink et al., who developed an environmental information system to support the exploration and visualization of energy systems, including subsurface storage and integration of renewable energy. Their framework provides a 3D virtual geographic environment for analyzing heterogeneous data, such as GIS layers, simulation results, and temporal datasets [4].

Graebler et al. present a Virtual Experiment Information System for the Mont Terri Underground Research Laboratory [1]. It also integrates heterogeneous datasets, such as simulation results and sensor data, into an interactive 4D visualization to explore geoscientific processes in radioactive waste repositories. The focus is on reducing cognitive and technical barriers, supporting collaboration, and enabling public science communication.

Unlike previous works, our Immersive Engineering Toolkit for deep geothermal energy uses an open-source VR engine to offer stakeholders a comprehensive framework for collaboration across the entire geothermal lifecycle. By bridging the gap between immersive scientific visualization and engineering applications, the toolkit enables the integration of heterogeneous data into a holistic approach. This supports informed decision-making, scenario analysis, risk assessment, staff training, and public participation. The toolkit is continuously developed and validated through ongoing deep geothermal projects, ensuring its relevance to real-world applications. By fostering a shared understanding of technical and environmental aspects, our approach aims to reduce costs, improve efficiency and safety, and strengthen research as well as stakeholder collaboration and engagement.

2 IMPLEMENTATION

The toolkit is implemented using an open source virtual reality and engineering framework PolyVR, which supports a wide range of immersive systems (e.g., CAVE, VR headsets), interaction devices, and operating systems [2]. The built-in collaboration module of the framework enables remote interaction between stakeholders, making it suitable for multi-user and distributed environments. Built-in simulation modules for physics and engineering systems, and data interfaces enable the simulation of machine behavior, based on CAD data and original machine programming. This facilitates the development of virtual twins of drilling equipment and machines,

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Category	Usage Types	Data Formats
Vector Data	Municipal boundaries, street maps, POIs	Shape, GeoJSON, CityGML, OSM
Raster Data	Satellite imagery, elevation models, bore core scan textures	SRTM, PNG, JPEG, TIFF
Spatial Data	Point clouds	xyz, e57
Scientific Data	Volumetric rendering, CT-Scans	VTK, DICOM
3D Assets	Machinery, construction, 3D models, height layers	STEP, Collada, GLTF, VRML, GOCAD ts
2D Assets	Planning layouts	DWG, DXF, PDF
Live Interfaces	Co-simulation, SiL, HiL, communication	REST, MQTT, OPC-UA, S7, UDP, TCP

Table 1: Integrated data types

especially for software and hardware in the loop, training simulations, realistic scenario building, and risk assessment.

To address the specific requirements of the geothermal domain, various data interfaces were developed (see 1). These include geographically referenced large-scale datasets, such as municipal boundaries, surface elevation models and subsurface surveys, as well as smaller-scale data, like photogrammetry and micro-CT scans of borehole core samples. The VR environment enables interactive visualization and analysis of these datasets, simplifying complex spatial and temporal interpretations.

2.1 Drill Site Planning on Surface and Subsurface

The drill site configurator (Figure 1) facilitates the planning and visualization of above-ground processes during the construction phase of a geothermal drilling site. Users can import 2D construction plans or layouts in common formats (e.g., dxf, dwg, pdf) and place site assets (i.e. drill rig, pipes, containers), using a UI with built-in configuration logic. This virtual environment enables users to interactively optimize layouts directly in the VR environment to enhance efficiency, safety training, and familiarize workers and stakeholders with the facilities. Regardless of the physical state of the drill site potential issues can be addressed at an early stage. The underground visualization and an integrated casing simulation display geological information of the underground structure as well as the process of the drilling and casing within the geothermal borehole (Figure 1). The geological layers were explored with 3D seismic imaging, processed in Petrel, and imported into the toolkit via GOCAD .ts format.

2.2 Public Engagement Applications

Two VR applications developed with the proposed toolkit focus on engaging the public: an immersive underground elevator and a visualization of a heat storage cycle. The elevator simulation allows users to virtually descend through the geological layers alongside the borehole, displaying depth indicators, temperature gradients, and key geological features. It incorporates textured visuals, mini-map for orientation, and voice output to explain the geology characteristics as well as safety measures such as prevention of ground-water contamination. The heat storage cycle application demonstrates the seasonal storage and retrieval of energy in deep geothermal systems through an interactive and easily understandable visualization. By providing immersive and scientifically accurate experiences, these applications help bridge knowledge gaps, fostering public understanding and trust in geothermal energy.

2.3 Tunnel Boring Machine Hybrid Twin

The hybrid twin of tunnel boring machine(TBM) simulator combines software and hardware simultaneously (Figure 1). The virtual machine can be steered from a physical control panel and runs in tandem with the programmable logic controller (PLC) of the TBM through real-time communication (OPC UA). This enables scenarios with real machine behaviour and fault states of machine code on the PLC [3]. The hybrid twin is used to train operators, can be applied for (co)simulation and validation of the boring process.

3 DISCUSSION

Stakeholders, such as permitting authorities responsible for water rights and environmental aspects, often lack experience in evaluating deep drilling projects and the associated risks posed by drilling and operations at the drill site. This often results in a very lengthy approval process. By providing a clear and illustrative representation of the subsurface conditions and associated risks, uncertainties and apprehensions caused by lack of knowledge can be mitigated. The same applies to the concerns of local residents and citizen initiatives. Another key aspect is the integration of the various engineering firms involved in construction, media supply and infrastructure. A streamlined management system that offers the described toolkit can help optimize on-site processes, thereby reducing expensive construction time.

4 CONCLUSION AND FUTURE WORK

Our Immersive Engineering Toolkit aims to unify diverse data and engineering processes in geothermal energy into a cohesive framework, providing intuitive tools to support research and industry.

To validate the usability of the toolkit, the next steps involve leveraging the immersive applications to discuss different matters with various stakeholders, such as permitting authorities and local residents. This includes events to inform citizens about geothermal heat storage potentials in their city and the installation of VR demonstrators at the information center at the drilling site, providing the public with an interactive and educational experience. Future works will focus on extending the toolkit to meet evolving engineering and digital twin requirements. A HiL simulation for vertical drill rigs should improve drill operations training processes, enable detailed simulations under varying geological conditions, and contribute to reducing drilling costs.

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